MIL-D-14130A
26 February 1986
SUPERSEDING
MIL-D-14130(CmlC)
3 August 1955

MILITARY SPECIFICATION

b,b'-DIHYDROXYDIETHYL SULFIDE (THIODIGLYCOL), TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of b,b'-dihydroxydiethyl sulfide which is also known as thiodiglycol.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-96 - Cans, Metal, 28 Gage and Lighter
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons)

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research and Development Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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STANDARDS

FEDERAL

Fed. Std. No. 313 - Material Safety Data Sheets Preparation and the Submission Of

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are indicated as DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM STANDARDS

D 891 - Specific Gravity, Apparent, Of Liquid Industrial Chemicals (DOD Adopted)
D 1193 - Reagent Water (DOD Adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Appearance. b,b'-Dihydroxydiethyl sulfide shall have a color no darker than light greenish yellow and shall be substantially free from suspended matter when tested as specified in 4.2.4.1

3.2 Chemical and physical characteristics. b,b'-Dihydroxydiethyl sulfide shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
b,b'-Dihydroxydiethyl sulfide, percent by weight	96.0	---	4.2.4.2
Specific gravity at 20°/20°C	1.181	1.186	4.2.4.3
Acidity, mg KOH per gram of specimen	---	1.0	4.2.4.4
Alkalinity, mg H_2SO_4 per gram of specimen	---	1.0	4.2.4.5
Mercaptans (as mercaptoethanol), percent by weight	---	0.50	4.2.4.6
Water, percent by weight	---	1.0	4.2.4.7
Ash, percent by weight	---	0.02	4.2.4.8

3.3 Material Safety Data Sheets. Material Safety Data Sheets for b,b'-dihydroxydiethyl sulfide shall be prepared and submitted by the contractor as specified in Fed. Std. No. 313 (see 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not

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relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the b,b'-dihydroxydiethyl sulfide, produced by one manufacturer in no more than 24 consecutive hours, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For b,b'-dihydroxydiethyl sulfide test. Sampling shall be conducted in accordance with table II. For 100-gram (g) bottles, the entire contents of the sample bottles shall be composited. For other containers, a representative specimen of approximately 300 g shall be removed from each sample container and placed in a suitable, dry, sealed container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for b,b'-dihydroxydiethyl sulfide test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking

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(e) Fiberboard partitions or pads missing or not as specified (where required)
(f) Marking incorrect, missing, or illegible

4.2.3.2 For b,b'-dihydroxydiethyl sulfide test. For 100-g bottles, the composited specimen from 4.2.2.2 shall be tested as specified in 4.2.4. For other containers, each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by the composited specimen or by any individual specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for color and suspended matter.

4.2.4.2 b,b'-Dihydroxydiethyl sulfide. Weigh to the nearest milligram (mg) approximately 100 g of specimen into a tared, 125-milliliter (mL) Claissen flask containing an internal ebullator. Attach the flask through a 500-millimeter (mm), water-cooled West condenser to a 125-mL distillation flask connected to a coldtrap and vacuum system. Fit the side neck of the Claissen flask with a thermometer of suitable range, calibrated for 100-mm immersion, graduated in 0.2°C units, and capable of being easily read to 0.1°C. Reduce the pressure on the system to 3 or 4 mm of mercury and apply heat by means of an oil bath. Heat the oil to 140°C and maintain the bath temperature at 140° to 145°C for 15 minutes. Observe the vapor temperature of the specimen; this should be approximately 40°C at the end of the heating period. Increase the temperature of the bath to 150°C; at this point the vapor temperature should rise quickly. Immediately release the vacuum and remove the Claissen flask. Clean the oil from the outside of the flask and reweigh to the nearest milligram. The loss in weight is a measure of the volatile components. Again connect the Claissen flask to the condenser, adjust the vacuum to 3 to 5 mm of mercury pressure, and heat the oil bath in such a manner that the specimen distills at a rate of one drop per second. Near the end of the distillation, maintain the temperature of the oil bath at 200° to 210°C until no more material distills. Release the vacuum, remove the Claissen flask, and reweigh to the nearest milligram. Record the weight of the remainder of the specimen in the Claissen flask as grams of residue. Calculate the percent by weight b,b'-dihydroxydiethyl sulfide as follows:

$$\text{Percent b,b'-dihydroxydiethyl sulfide} = \frac{100(W - A - B)}{W}$$

where: A = Weight of volatile compounds in grams,
B = Weight of residue in grams, and
W = Weight of specimen in grams.

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4.2.4.3 Specific gravity. Determine the specific gravity of the specimen at 20°/20°C in accordance with the pycnometer method of ASTM D 891.

4.2.4.4 Acidity. Transfer 100 mL of water to a 250-mL Erlenmeyer flask. Add a few drops of a 1-percent alcoholic solution of phenolphthalein and neutralize with standard 0.1N alcoholic potassium hydroxide solution. Pipet 25 mL of specimen into the flask and titrate with the standard 0.1N alcoholic potassium hydroxide solution to a pink end point which lasts for at least 15 seconds. Calculate the acidity, as milligrams KOH per gram of specimen, as follows:

$$\text{Acidity} = \frac{2.24AB}{W}$$

where: A = Milliliters of alcoholic potassium hydroxide solution used to titrate the specimen,
B = Normality of the alcoholic potassium hydroxide solution, and
W = Specific gravity of the specimen determined in 4.2.4.3.

4.2.4.5 Alkalinity. Transfer 100 mL of water to a 250-mL Erlenmeyer flask. Add a few drops of a 0.05-percent alcoholic solution of methyl orange and neutralize with standard 0.1N sulfuric acid. Pipet 25 mL of specimen into the flask and titrate with the standard 0.1N sulfuric acid to an orange-pink end point which lasts for at least 15 seconds. Calculate the alkalinity, as milligrams H_2SO_4 per gram of specimen, as follows:

$$\text{Alkalinity} = \frac{1.96AB}{C}$$

where: A = Milliliters of sulfuric acid used to titrate the specimen,
B = Normality of the sulfuric acid, and
C = Specific gravity of the specimen determined in 4.2.4.3.

4.2.4.6 Mercaptans. Transfer 100 mL of water to a 250-mL Erlenmeyer flask. Add 8.0 to 10.0 g of the specimen weighed to the nearest 0.1 mg by means of a weighing pipet. Add 3 mL of 1-percent aqueous starch solution and titrate with standard 0.1N iodine solution to a blue end point which lasts for at least 10 seconds. Calculate the percent by weight mercaptans as mercaptoethanol as follows:

$$\text{Percent mercaptans} = \frac{7.81AB}{W}$$

where: A = Milliliters of iodine solution used in the titration,
B = Normality of the iodine solution, and
W = Weight of specimen in grams.

4.2.4.7 Water. Transfer 25 mL of anhydrous methanol to each of two dry, 125-mL, glass-stoppered Erlenmeyer flasks. Titrate the contents of each flask with Karl Fischer reagent to the same light reddish-brown color. Reserve one

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of the flasks as a blank. Into the second flask add 4.0 to 5.0 g of specimen weighed to nearest 0.1 mg by means of a weighing pipet. Titrate immediately with Karl Fischer reagent until the color matches that of the blank. Calculate the percent by weight water as follows:

$$\text{Percent water} = \frac{AB}{W} - 0.231C$$

where: A = Milliliters of Karl Fischer reagent required for specimen,
B = Grams of water equivalent to 100 mL of Karl Fischer reagent,
C = Percent by weight mercaptans determined in 4.2.4.6, and
W = Weight of specimen in grams.

4.2.4.8 Ash. Weigh to the nearest 0.1 mg approximately 50 g of specimen into a 125-mL platinum dish which has been ignited to constant weight, cooled in a desiccator, and tared to the nearest 0.1 mg. Heat the dish until the vapors continue to burn after the flame is withdrawn. Protect the combustion from drafts, and allow the vapors to burn spontaneously. Ignite the dish to a dull red heat, allow to cool in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent by weight ash as follows:

$$\text{Percent ash} = \frac{100(A - B)}{W}$$

where: A = Weight of dish and residue in grams,
B = Weight of dish in grams, and
W = Weight of specimen in grams.

5. PACKAGING

5.1 Preservation, level A.

5.1.1 Unit packing. b,b'-Dihydroxydiethyl sulfide shall be unit packed in a 100-g bottle, 1-gallon (gal) can, 5-gal pail, or 55-gal drum as specified (see 6.2).

5.1.1.1 One hundred-g bottle. A quantity of 100 (+1 or -0) g of b,b'-dihydroxydiethyl sulfide shall be unit packed in a glass bottle conforming to group A, class 1, type d, style 1, grade 1, closure A, seal A of PPP-C-186.

5.1.1.2 One-gal can. A quantity of 9.5 (+0.1 or -0) pounds of b,b'-dihydroxydiethyl sulfide shall be unit packed in a 1-gal tinplate or terneplate can conforming to type V, class 2 of PPP-C-96.

5.1.1.3 Five-gal pail. A quantity of 45.0 (+0.5 or -0) pounds of b,b'-dihydroxydiethyl sulfide shall be unit packed in a 5-gal steel pail conforming to type I, class 3 of PPP-P-704.

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5.1.1.4 Fifty-five-gal drum. A quantity of 530 (+5 or -0) pounds of b,b'-dihydroxydiethyl sulfide shall be unit packed in a 55-gal drum conforming to type II of PPP-D-729.

5.1.2 Intermediate packing.

5.1.2.1 One hundred-g bottle. Ninety-six bottles unit packed as specified in 5.1.1.1 shall be intermediately packed in a box conforming to style RSC, grade W5c, class weather-resistant of PPP-B-636. The bottles shall be arranged in two layers in a six by eight pattern with snug-fitting, interlocking partitions and full top, bottom, and intermediate pads. Partitions and pads shall be constructed of fiberboard conforming to grade W6c, class weather-resistant of PPP-F-320. The box shall be closed in accordance with method III followed by method V of the appendix to PPP-B-636.

5.1.2.2 One-gal can. Six cans unit packed as specified in 5.1.1.2 shall be intermediately packed in box conforming to style RSC, grade V3c, class weather-resistant of PPP-B-636. The box shall be closed in accordance with method III of the appendix to PPP-B-636.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-hundred-g bottle. Ninety-six bottles intermediately packed as specified in 5.1.2.1 shall be packed in a nailed wood box conforming to class 2, style 4 of PPP-B-621.

5.2.1.2 One-gal can. Six cans intermediately packed as specified in 5.1.2.2 shall be packed in a nailed wood box conforming to class 2, style 4 of PPP-B-621.

5.2.1.3 Five-gal pail and 55-gal drum. The unit pack shall serve as the shippng container.

5.2.2 Level B.

5.2.2.1 One hundred-g bottle and 1-gal can. Packing shall be as specified for level A packing except that the box shall conform to class 1 of PPP-B-621.

5.2.2.2 Five-gal pail and 55-gal drum. The unit pack shall serve as the shipping container.

5.2.3 Level C.

5.2.3.1 One hundred-g bottle and 1-gal can. The intermediate packing container specified in 5.1.2.1 or 5.1.2.2, as applicable, shall serve as the shipping container.

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5.2.3.2 Five-gal pail and 55-gal drum. The unit pack shall serve as the shipping container.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. In addition, each unit container shall be marked to show the following information:

WARNING!

CAUSES IRRITATION

Avoid contact with eyes, skin, and clothing.
Wash thoroughly after handling.

FIRST AID: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Call a physician. Flush skin with water. Wash clothing before re-use.

6. NOTES

6.1 Intended use. b,b'-Dihydroxydiethyl sulfide is intended for use in the manufacture of chemicals.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Unit quantity required (see 5.1.1)
(c) Level of packing required (see 5.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.5 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with Fed. Std. No. 313. The pertinent mailing addresses for submission of data are listed in appendix B of Fed. Std. No. 313.

Custodian:

Army - EA

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-14130A	b,b'-DIHYDROXYDIETHYL SULFIDE (TRIODIGLYCOL), TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify):* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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